

SOP Basics

SoP should highlight:

- 1- Area of interest (& possibly profs you want to work with)
- 2- Briefly: your academics (grades, honors, projects)
- 3- Your prior and ongoing research projects. For each proj, talk about motivation, your role & contribution, & outcome/status

Berkely manual: <https://grad.berkeley.edu/admissions/steps-to-apply/requirements/statement-purpose/>

Tips:

1. What the admissions committee will read between the lines: self-motivation, competence, potential as a graduate student.
2. Emphasize everything positively and write in an active, not a passive voice.
3. Demonstrate everything by example; don't say directly that you're a persistent person, show it.
4. If there is something important that happened to you that affected your grades, such as poverty, illness, or excessive work, state it. Write it affirmatively, showing your perseverance despite obstacles. You can elaborate more in your personal statement.
5. Make sure everything is linked with continuity and focus.
6. Unless the specific program says otherwise, be concise; an ideal essay should say everything it needs to with brevity. Approximately 500 to 1000 well-selected words (1-2 single space pages in 12 point font) is better than more words with less clarity and poor organization

SOP

Research experience

Currently (Xulab)

- Currently involved in research in Xulab doing research in computer vision applications in biomedical image analysis
- The research is focused on Cryo Electron tomography. Specifically, it involves applying 3D computer vision techniques on tomograms to detect, segment, recognize, and pick different kinds of biological particles from the tomograms.
- My current focus is developing new particle-picking techniques based on deep learning for specific tomogram datasets, EMPIAR 1007.
- This involves the use of 3D computer vision techniques.

Summer (Federated GML)

- During the summer, I worked on federated graph machine-learning tasks. Specifically, my research focused on developing techniques for federated learning graph neural networks. The goal was to develop approaches to ensure minimal degradation in the performance of GNN training and when training different GNN models with Graph data. As a result, I developed a computationally efficient approach that allows for training the different GNN models with clients in more than ten nodes. Current results indicate that even without any inter-client communication, we can reduce performance degradation from 20 % to less than 5% even when we have ten clients.

Safety critical scenario generation

- As part of a class, this research project involved the generation of safety-critical scenarios for self-driving vehicles—the research aimed to generate synthetic data that can be used in training and evaluating AD planners. The safety-critical scenarios are usually underrepresented in AD datasets, and this may be a problem since the planners may not be able to function as expected when exposed to these safety-critical scenarios. In this research, we made some improvements, specifically in trajectory modeling, by exploring various ways to model the trajectories of ego vehicles in the traffic scene. The result so far was an improvement in the realism and utility of these safety-critical scenarios.

Depth Estimation using Midas & Yolo:

- For the computer vision class, I applied Yolo & Midas models to perform depth estimation

Research goals

My research goals involve developing AI, deep learning, and machine learning solutions that impact people's lives, especially in the global south. I am particularly interested in using various AI techniques in healthcare, education, transport, and engineering. With a strong background in engineering and multiple disciplines, I want to direct my research efforts to areas that can impact people's lives. I have previously done this by applying AI in bio, medical image analysis, and safety-critical scenario generation for transport applications. I am also using computer vision to explore the safety and efficiency of traffic scenes in Kigali.

Describe projects

Draft 1

Part 1: Introduction, motivation & interests

I am Brian Kipkirui, a final year master's student at Carnegie Mellon University Africa, pursuing a master's in artificial intelligence engineering. Before that, I completed a Double major undergraduate degree in Electrical & Electronics Engineering and Electronics and computer engineering. Over the past year, I have taken foundational AI and machine learning courses, including Deep learning, Machine learning, computer vision & NLP. In addition, as a research intern, I have had the chance to work on class projects and research projects. In addition to these, I have had the opportunity to work as a teaching assistant for a course on the mathematical foundations of machine learning. This experience has allowed me to learn and understand foundational concepts in AI and machine learning, develop technical skills in deep learning and machine learning, gain research experience, and develop other skills through organization, mentorship, and administrative responsibilities as a teaching assistant. My goal is to improve my research skills, dive deeper into Machine learning and AI, and do more research in these areas in ways that can be applied to tackle real work problems, specifically in the developing world.

Part 2: Summary of previous graduate and Undergraduate career

During the summer, I worked on federated graph machine-learning tasks. Specifically, my research focused on developing techniques for federated learning graph neural networks. The goal was to develop approaches to ensure minimal degradation in the performance of GNN training and when training different GNN models with Graph data. As a result, I created a computationally efficient approach that allows for teaching different GNN models to clients in more than ten nodes. Current results indicate that even without inter-client communication, we can reduce performance degradation from 20 % to less than 5% even when we have ten clients. I plan to submit a first-authored paper to MLSys 2025 by November 2024.

In addition to this, as part of a class, this research project involved the generation of safety-critical scenarios for self-driving vehicles—the research aimed to generate synthetic data that can be used in training and evaluating AD planners. The safety-critical scenarios are usually underrepresented in AD datasets, which may be a problem since the planners may need to function as expected when exposed to these safety-critical scenarios. In this research, we made some improvements, specifically in trajectory modeling, by exploring various ways to model the trajectories of ego vehicles in the traffic scene. The result so far was an improvement in the realism and utility of these safety-critical scenarios. I plan on building on this research to create datasets and analyze various traffic scenarios in Kigali for safety and efficiency.

Part 3: Recent and current activities

I am currently involved in research in Xulab, doing research in computer vision applications and biomedical image analysis. The research is focused on Cyro Electron tomography. It consists of applying 3D computer vision techniques on tomograms to detect, segment, recognize, and pick different kinds of biological particles from the tomograms. My current focus is developing new particle-picking methods based on deep learning for specific tomogram datasets, EMPIAR 1007. This involves the use of 3D computer vision techniques. This and other research activities have allowed me to develop my research

skills and, more importantly, apply deep learning and computer vision techniques in solving real-world problems.

Part 4: Elaborate on academic interests

Biology/Medical imaging

GNNs in biology for molecules

Computer vision for medical imaging,

CryoET

NLP in ...

Material Science

NLP for information retrieval and making up recipes

GNNs

Computer vision for robotics and synthesis of recipes

Transport & mobility

Analysis of traffic scenarios for safety and efficiency

Analyzing traffic footage for proper urban design

Evaluating AD planners and perception modules for roads in Africa

All these learning, skills, and experience gained from my coursework and research have provided me with an excellent foundation to further my studies in research. My current academic interests include researching AI and deep learning in computer vision, robotics, and NLP, and I am applying them in different fields in science and engineering. These include the application of computer vision in biomedical image analysis, GNNs in studying protein and other molecular structures, and material science. I am also excited about applying these, including NLP computer vision and GNNs, in transport and material science.

GPT Version of Draft 1

Statement of Purpose

Introduction and Motivation

I am Brian Kipkirui, a final-year Master's student in Artificial Intelligence Engineering at Carnegie Mellon University Africa. My academic journey has been deeply rooted in the pursuit of using AI to address real-world problems, particularly in underrepresented areas such as healthcare and transportation. My undergraduate studies in Electrical and Electronics Engineering, paired with my strong foundation in AI and machine learning, have allowed me to explore the potential of AI to drive societal impact. My goal is to further this pursuit through advanced research that applies cutting-edge machine-learning techniques to areas like biomedical imaging, autonomous systems, and federated learning, particularly in ways that can benefit developing regions.

Research Experience

Over the course of my Master's program, I have been fortunate to engage in several research projects that have allowed me to deepen my expertise in machine learning and its applications across multiple domains.

Currently, I am conducting research at Xulab, where I am exploring computer vision applications in biomedical image analysis, specifically Cryo-Electron Tomography (CryoET). This research focuses on applying 3D computer vision techniques to detect, segment, and classify biological particles within tomograms. My current project centers on developing deep learning-based particle-picking techniques for cryoET datasets, particularly EMPIAR 1007. This work is significant as it advances the automation of complex biomedical imaging tasks, which is critical for applications like structural biology and drug discovery.

During the summer, I worked on federated learning techniques for Graph Neural Networks (GNNs). My research aimed to address the performance degradation typically associated with training GNN models in federated settings. I developed a computationally efficient approach allowing decentralized training across multiple clients, achieving performance degradation of less than 5% across ten nodes without inter-client communication, compared to the initial 20%. This approach enhances the scalability and applicability of federated learning for GNNs, which are essential for tasks such as social network analysis and molecular graph studies. I plan to submit my findings as a first-authored paper to MLSys 2025 by November 2024.

In another project, I contributed to the generation of safety-critical scenarios for autonomous driving systems. This research, conducted as part of a course, focused on creating synthetic data to train and evaluate autonomous driving (AD) planners in safety-critical situations. The rarity of such scenarios in existing datasets poses a challenge for AD systems in real-world conditions. By improving trajectory modeling for ego vehicles, my work contributed to the generation of more realistic and diverse safety-critical scenarios, which in turn improves the robustness of AD planners. I intend to extend this work by analyzing traffic scenes in Kigali, Rwanda, to explore how AI can enhance safety and efficiency in urban mobility.

Academic Background and Skills

My academic coursework has provided a solid foundation in AI, deep learning, and machine learning, covering topics such as computer vision, NLP, and reinforcement learning. As a research intern and teaching assistant at Carnegie Mellon, I have developed both technical and mentorship skills while deepening my understanding of foundational AI concepts. As a teaching assistant for the Mathematical Foundations of Machine Learning course, I honed my ability to explain complex concepts clearly and provided support to students, which has greatly strengthened my communication skills.

Research Goals and Academic Interests

My research interests lie at the intersection of AI, deep learning, and their applications to critical areas such as healthcare, transportation, and material science. Specifically, I aim to explore:

Academic Interests 1:

My research interests lie at the intersection of AI, deep learning, and healthcare applications, particularly in computational biology and biomedical image analysis. I am passionate about leveraging advanced computer vision and graph neural networks (GNNs) to tackle complex medical problems, where data is often sparse and highly structured. My current work focuses on developing deep learning methods for Cryo Electron Tomography (CryoET) to analyze molecular structures, such as in the EMPIAR 1007 dataset. I also see potential in applying GNNs to model intricate biological networks, such as protein interactions, to advance drug discovery. Additionally, I aim to explore the use of AI for broader medical imaging challenges, from radiology to histopathology, to improve diagnostic accuracy and patient outcomes. I am excited to further this research in a PhD program, particularly at [University], where the emphasis on AI-driven healthcare solutions aligns with my long-term goal of advancing medical innovations through machine learning.

Academic Interests 2:

My research interests also extend to applying AI, deep learning, and computer vision in traffic analysis, mobility, and autonomous driving. I have worked on generating safety-critical scenarios for autonomous vehicle planners, improving trajectory modeling to represent real-world traffic dynamics better. By applying computer vision techniques to traffic footage, I aim to enhance the safety and efficiency of urban mobility systems, particularly in developing cities like Kigali. Additionally, I see significant potential in using NLP and AI to analyze traffic patterns, optimize route planning, and improve the interaction between autonomous vehicles and human drivers. These advancements can lead to safer and more efficient transportation systems globally. I am eager to explore these applications further in a PhD program, particularly at [University], where innovative AI research drives the future of autonomous systems and intelligent transportation.

Academic interests 3:

With a solid foundation in AI, deep learning, and computer vision, my research experience has spanned diverse applications, from biomedical imaging to federated learning for graph neural networks (GNNs). I am particularly excited to extend these skills to material science, where AI-driven methods can accelerate the discovery and synthesis of new materials. Techniques like GNNs are crucial for modeling molecular structures and predicting material properties, while computer vision can be leveraged to analyze microscopic material images for defect detection and quality control. I believe that applying these AI tools to material research can significantly impact energy materials, sustainable building materials, and

healthcare-related nanomaterials—challenges that are particularly pressing in Africa and the developing world. By optimizing material discovery and synthesis pathways through AI, we can develop affordable and sustainable solutions to meet the unique needs of developing regions, addressing issues like renewable energy storage, construction, and medical devices.

Federated Learning for GNNs: Advancing decentralized learning methods for scalable graph-based systems, with applications in healthcare, drug discovery, and social networks.

Autonomous Systems and Urban Mobility: Using AI to enhance safety-critical scenario generation and improve traffic efficiency in developing cities, such as Kigali.

Material Science and AI: Applying NLP and GNNs to accelerate research in material discovery, focusing on synthesizing new materials for energy and healthcare applications.

I am excited by the prospect of furthering my research in these areas through Carnegie Mellon's PhD program in Computer Science. I am particularly drawn to the work of professors such as [Professor X] and [Professor Y], whose research in federated learning and AI-driven healthcare aligns with my goals. Carnegie Mellon's collaborative research environment, strong industry ties, and focus on AI for social impact make it the ideal place for me to contribute to cutting-edge research while developing scalable solutions for real-world challenges.

Conclusion

In summary, my academic background, research experiences, and passion for applying AI to solve global challenges have prepared me to thrive in Carnegie Mellon's PhD program. I am eager to deepen my research skills, collaborate with world-class researchers, and contribute to advancing AI in impactful ways, particularly in healthcare and autonomous systems. With a clear vision and strong motivation, I am confident that this program will allow me to make meaningful contributions to the field and society.

Draft 2:

Here's a refined Statement of Purpose that is general and adaptable for PhD applications in Computer Science programs specializing in AI, NLP, ML, and Computer Vision:

****Statement of Purpose****

****Introduction and Motivation****

My name is Brian Kipkirui, and I am a final-year Master's student in Artificial Intelligence Engineering. Throughout my academic and professional journey, I have been driven by the potential of AI to address pressing societal challenges, particularly within healthcare, transportation, and material science. With an undergraduate foundation in Electrical and Electronics Engineering and advanced training in machine

learning and computer vision, I am eager to push the boundaries of AI to create tangible, positive impacts on global and regional scales. Pursuing a PhD in Computer Science will allow me to develop and apply cutting-edge AI techniques to solve complex problems, building on my research experience and my commitment to innovation for social good.

****Research Experience and Accomplishments****

My research experiences span AI applications in biomedical imaging, autonomous systems, and federated learning, equipping me with both technical expertise and a deep understanding of these fields' real-world implications. Below are three projects that illustrate my skills, experience, and research potential.

* ****Biomedical Image Analysis with Computer Vision:**** Currently, I am engaged in research on Cryo-Electron Tomography (CryoET), focusing on particle detection, segmentation, and classification within tomograms. This work requires the development of 3D computer vision techniques to automate image analysis tasks within complex biological structures. My project specifically involves creating deep learning-based particle-picking methods tailored to datasets like EMPIAR 1007. This experience has honed my skills in 3D computer vision while deepening my understanding of its critical applications in structural biology and drug discovery.

* ****Federated Learning for Graph Neural Networks:**** As a Graduate Research Intern, I conducted research in federated learning for Graph Neural Networks (GNNs), devising a new method to improve training efficiency while reducing communication overhead. This approach demonstrated significant improvement, achieving less than 5% performance degradation across multiple client nodes without inter-client communication. The success of this project has furthered my interest in decentralized AI, and I am preparing a first-authored paper based on this research for submission. This experience underscored the potential for scalable federated learning models in sectors like healthcare, drug discovery, and social network analysis.

* ****Safety-Critical Scenario Generation for Autonomous Vehicles:**** In a collaborative project on autonomous driving, I focused on generating synthetic safety-critical scenarios for autonomous driving systems. My work centered on improving trajectory modeling for ego vehicles, contributing to more realistic and diverse safety-critical scenarios to better train autonomous vehicle planners. This experience exposed me to the challenges of ensuring safety and robustness in autonomous systems and inspired a broader interest in exploring AI applications in transportation and urban mobility.

****Research Interests and Future Directions****

My research interests lie at the intersection of AI, deep learning, and applications that span healthcare, transportation, and material science. In my PhD research, I aim to explore the following areas:

* ****Biomedical Imaging and Computational Biology:**** Building on my experience in CryoET, I am deeply interested in using computer vision and GNNs for biomedical imaging and computational biology. I envision using these tools to support advancements in drug discovery, disease diagnosis, and

personalized medicine. Additionally, I hope to explore AI applications for other medical imaging modalities, such as radiology and histopathology, to improve diagnostic precision and patient care.

* **Autonomous Systems and Urban Mobility:** My experience in safety-critical scenario generation has led to a strong interest in AI applications for urban mobility and transportation systems. By applying computer vision and deep learning, I aim to develop models that enhance the safety and efficiency of autonomous driving systems. I am particularly interested in studying the unique challenges of transportation systems in developing regions, where infrastructure and traffic patterns often differ from those in more developed settings. Through this research, I hope to contribute to safer and more efficient urban mobility solutions.

* **Material Science and AI-Driven Discovery:** AI, especially GNNs and computer vision, holds transformative potential for material science. I am intrigued by the opportunity to apply AI to accelerate the discovery and synthesis of new materials for energy storage, sustainable building materials, and healthcare applications. By leveraging GNNs for molecular structure modeling and computer vision for quality analysis, I aim to help meet the critical needs of developing regions through innovative materials that are both affordable and sustainable.

Alignment with PhD Program Goals

The convergence of my research experience in AI, computer vision, and GNNs, along with my passion for addressing real-world problems, aligns closely with the objectives of a PhD in Computer Science. I am especially drawn to programs that emphasize collaborative research, interdisciplinary applications, and a focus on AI-driven solutions for societal impact. I am eager to contribute to a vibrant research community, working with leading experts in my field while deepening my knowledge of foundational AI concepts.

Conclusion

In summary, my academic and research experiences have provided me with a solid foundation in AI and its diverse applications. I am excited to bring my skills and determination to a PhD program, where I can further advance my research capabilities, contribute to meaningful projects, and develop solutions that drive positive societal change. I am confident that my passion, technical skills, and commitment to impactful research make me a strong candidate for a PhD in Computer Science, and I look forward to the opportunity to contribute to the field and make a difference through AI.

This SoP is tailored to showcase your research strengths and interests while remaining adaptable to any top Computer Science PhD program specializing in AI, ML, NLP, and Computer Vision.